

The objective of this report is to analyze the AirBB Dataset provided, which consists of 74,111 observations of properties in 6 cities of the US, bookable on the AirBB platform. Our analysis brings insights into how the variables in the dataset are correlated with each other and then guides AirBnB landlords and customers. As the dataset data consists of 26 variables, we decided to focus on the variables that proved to be most insightful, or so those with the greatest correlations with Price, Reviews, and Ratings. We assumed that the Number of Reviews and the Ratings were good indicators of how visited and appreciated by customers each property is.

We begin with an overview of the main variables. The mean Price of all observations is 160.343 overall and 55.327 per person. Instead, the average Number of Reviews is 20.89(SD=37.782), with an average rating of 94.07(SD=7.834). While Log Prices are symmetric around the median¹, the Number of Reviews is right-skewed² (mean=20.9, median=6.0) and the Score Rating is left-skewed³(mean=94.07, median=96), portrayed by both the boxplots and the histograms. This depicts that properties have equally distributed rental prices; most do not have many reviews, but very often have high Review Score Ratings.

Property type and Room Type are structural elements of the rentals. They are both categorical nominal variables. Property Type analyzes 35 different types of properties with a minimal relevant number of observations, especially as 66.2% of properties are Apartments. The Room Type is more significant: there are three data points, Entire Home/Apt(55.7%), Private room(41.3%), and Shared room(3%). These percentages slightly vary depending on the city. As shown in the box plot⁴ and summaries, there is a strong correlation between Log Prices and Room type: Entire Home/Apt(Mean Log Price = 5.167) are more expensive than Private Rooms(4.328) which are in turn more expensive than Shared Rooms(3.862). This is possibly due to the increased number of accommodates in Entire Homes(Mean=4.079) compared to Private Rooms(1.988) and shared rooms(2.044). It is similar for the Number of Reviews and Review Score Ratings: the mean ratings for each Property Type are decreasing: Entire Home/Apt=94.302, Private Rooms=93.860 and Shared Rooms=91.872. Likewise for the Number of Reviews: Entire Home/apt(Mean=1.612), Private room(20.470) and Shared room(12.956). This could highlight the inclination of consumers to leave a review, as well as a better score rating if the rental is larger, and has more space. However, we also notice the larger the place, the more expensive the price tends to be. Hence, it seems that Entire Homes are more successful for landlords and, for customers, generally favoured. On the one hand, they might be more expensive and hence of better quality and more liked by consumers. On the other hand, Entire Homes might simply be more preferred and hence the increased demand drives up the price level.

Another crucial aspect of rental properties is their position, which in the dataset is encompassed by: the categorical variables City & Neighbourhood and the continuous variables Latitude & Longitude. In terms of the average rental, Log price SF(mean=5.170) costs the most, while Chicago the least (4.620). This is again evidenced by the boxplot⁵ on the R file. To further analyze this variable we created three absolute frequency bar plots: barplot1⁶ illustrates the observations in each city; barplot2⁷ the accommodations whose price is over the third quartile by city; and barplot3⁸ the accommodations with price under the first quartile again by city. In their comparison, it is again evidenced that San Francisco has a higher percentage of accommodations with a high price, while Chicago has a higher percentage of accommodations with a low price.

Moving forward, in terms of the Number of Reviews, and the corresponding Score Ratings: San Francisco has the most(mean=28.692), then Chicago(26.242), Boston(25.694), LA(21.260), DC(19.912) and lastly NYC(18.119). The most expensive and cheapest rental locations end up having the most reviews. It could possibly be due to the fact that when landlords' price differentiate themselves(most expensive or cheapest) consumers are more inclined to leave a review. This translates to the score ratings as well: on average Chicago(mean=95.101) has the best ratings followed by SF(95.020), DC(94.831), LA(94.302), Boston(93.600), and then NYC(93.507). These results highlight to customers the most expensive cities and their corresponding ratings, when planning a trip.

Diving deeper into properties' locations, how do the variables correlate inside each city? Comparing boxplots for each Neighbourhood⁹ is inconclusive since the number of observations for some categories is too small. Therefore, using the continuous variable Latitude and Longitude, we focused on broader areas. As an example, we analyzed NYC's five districts: Manhattan, Bronx, Brooklyn, Staten Island and Queens. Both average prices¹⁰ and Number of Reviews are lowest in the Bronx (namely mean = 4.252 and 14.73) and highest in Manhattan (namely mean = 4.850 and 18.12) whereas the average Review Score Rating only changes marginally in each district, highest in Brooklyn(93.70) and lowest in the Bronx(93.31). This highlights Manhattan being the most expensive rental area and the Bronx being more affordable but of lower quality. Is Brooklyn possibly the more balanced alternative? It does have the most positive review ratings, but the difference in ratings is not significant enough to draw strong conclusions.

Moving forward, onto the numerical count data variables. The number of Accommodates(Mean=3.155), Beds(1.711), Bedrooms(1.266) and Bathrooms(1.235) are all indicative of how many individuals the property can guest, and hence significant properties of any rental. Analyzing Accommodates, it is evident both in the box plot¹¹ and the scatter plot(+best-fit line)¹², that as Accommodates increases, the property's price increases. This is supported by the relatively significant correlation coefficient:

0.568. However, if instead, we take under observation the relationship between the number of Accommodates and the Log Price per Accommodate¹³, the expense isn't largely affected and remains fairly constant. Indeed, the correlation is -0.221, supporting our intuition. In fact, as Accommodates increase, the price per individual seems to slightly decrease. This is intuitive as every extra individual shares the same elements of the property. The other variables highlight this similar pattern. Rightfully so, as the number of Beds¹⁴, Bedrooms¹⁵ or Bathrooms¹⁶ increases, the Log Price also increases. This is supported by the boxplots and correlation values: 0.443, 0.473, and 0.355 correspondingly. Yet, once again, as soon as we compare these variables to the Price per Accommodate, there are less significant correlations: -0.166, -0.032, 0.027. These variables are all connected and have similar dynamics. They demonstrate that yes, clearly, the more individuals, the higher the price. However, if you add or remove an extra accommodate, in most cases the price per person will remain the same, if not decrease. What advice can we provide stakeholders: landlords should try to increase their number of Accommodates for higher revenue, instead customers should realize that two rentals for 3 individuals would cost the same per person as one for 6.

Amenities are another variable, of categorical nominal type, that can provide insight into the rentals' price dynamics. The variable consists of 87 possible different data points. The mean Amenities per property is 16.604(SD=6.935) and varies marginally(± 3) across cities. We divided the prices into three ranges (under the first quartile, interquartile range, and over the third quartile) and calculated the conditional probabilities for 9 chosen amenities. It is evident that higher prices are associated with a higher presence of these 9 amenities (both basic and more refined). This is consistent visually with the boxplot¹⁷ between the number of Amenities and Log Price, especially when considering only properties with less than 50 amenities(89% of the data). Staying in this same range of data it is also evident that, when increasing the number of Amenities, also the Number of Reviews increases. This could highlight to landlords that the more Amenities, the more reviews and the higher the Price point they can ask for.

The details of AirBnB properties, or so the binary variables: Cleaning fee, Host Identity Verified and Instant bookable also hold a significant role in positioning the rental. 73% of properties require a contribution for cleaning. These tend also to be the properties which are on average more expensive. The mean Log Price is 4.830 when 'True' compared to 4.649 when 'False'. The Number of Reviews is similar: the mean is 23.387 when 'True' vs 13.982 when 'False'. More reviews, but also better ratings: the mean when 'True' is 94.206 whereas when 'False' is 93.526. Possibly, the cleaning fee ensures a cleaner property for customers. The analysis highlights for landlords the importance of having a required cleaning contribution. They can gain a higher rental revenue as well as have more and better reviews. It also hints to customers to search for houses with a required cleaning fee, as they usually have a better rating.

The host's identity being verified or not, is a similar variable. When verified(in 67% of properties): the mean Log Price is slightly higher(4.794 vs 4.757), the mean Number of Reviews is significantly more(25.167 vs 12.097), and also the mean Score Rating is greater(94.333 vs 93.353). On the one hand, this is a clear indication for landlords to verify their accounts to receive a higher price alongside more and better reviews. A verified account instils more trust in any customer. On the other, customers should filter whether the landlord is verified or not and keep in mind that, for a better stay, verified accounts receive more and better reviews.

However, for the variable Instant Bookable, it is the opposite: the mean Log Price is greater if it is not bookable instantly(4.801 vs 4.729). This might be because more expensive rentals could require further processes or confirmation. Instead, in terms of ratings, it seems that when Instant Bookable = 'False' on average you receive fewer reviews(Mean=12.097 vs 25.167) but with better Score Ratings(Mean=94.391 vs 93.178). Therefore, in theory, landlords would benefit from not being instantly bookable. However, to what extent are the better ratings truly motivated by the bookability, and not the price? It could either be a correlation or indeed one causing the other.

The analysis was effective in highlighting to AirBnB landlords and customers relevant key factors that can affect the positioning of the rental property. It is of high value for readers as it handles a sample of data which spans more than 70 000 observations. Hence even minor differences in prices or changes in variables, provide reliable, meaningful and accurate correlations. The analysis is however limited. Many correlations were hindered upon for some categories offsetting the results due to having too few data points. More importantly, the exploration struggled in understanding whether some conclusions made were indeed due to causation or instead simply a correlation. Many of the variables move in a similar direction, so it is therefore difficult to truly grasp which is the cause for correlation. For example, how does one interpret the extent to which the price increases due to more Bathrooms, or instead more Accommodates? It was difficult to avoid this Omitted variable bias. As improvements, the investigation itself could expand upon these areas. Instead, the data sample provided could try to include other relevant data such as the property price or the rentals' ratio of occupation, to provide further economical insights for landlords specifically.